

SOFTWARE PROJECT PRESENTATION · 2026

GLASSWALL

A privacy-preserving browser agent with on-device visual perception

An AI agent that fills web forms without the AI ever seeing your personal data.

Web page

name, email, phone, card



On the device

read · detect · redact



Over the network

[[EMAIL#1]] [[PHONE#2]]



LLM planner

returns one action

Real values never leave the browser. The form still fills.

DESCRIPTION

What GLASSWALL is, and why

A Chrome extension plus a small server that let an AI model operate a web page for the user, without the model ever receiving the personal data on that page.

1

Reads the page on the device

structure, text and pixels

2

Replaces personal values with handles

an email becomes [[EMAIL#1]]

3

The AI plans, the device acts

the real value is filled in locally

THE PROBLEM TODAY

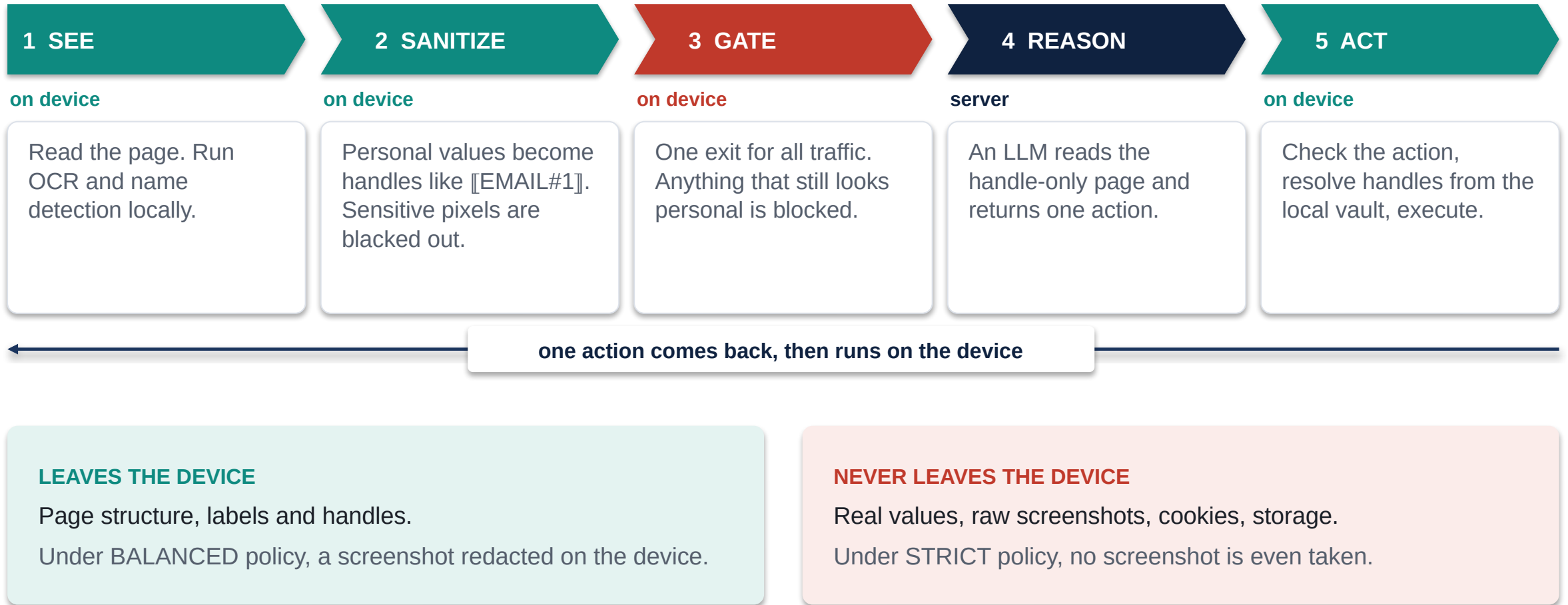
PATIENT ID	NAME	MRN	PHONE	DOB	ACTIONS
P001	Vivek Gupta	CLIN-300178	+917757420286	1998-10-01	View
P002	Vikram Patel	OPD-123733	+919882942278	1956-01-25	View
P003	Ajay Reddy	MRN-243892	+916424795596	1997-07-05	View

Browser agents upload the whole screen to a cloud model.

Anything the detector misses is sent as-is.

Hiding a value from the AI means it cannot fill the form.

One step of the agent loop



Already built and running

The screenshot shows a checkout page with the following elements and annotations:

- Checkout** (e13)
- Your saved details** (e15)
- Name** (e16): DROP: PERSON_NAME from ner+regex (S=0.98)
- Email** (e18): DROP: EMAIL from regex (S=0.99)
- Phone** (e20): DROP: PHONE from regex (S=1.00)
- Address** (e22): DROP: STREET_ADDRESS from ner+regex (S=0.98)
- City** (e24): DROP: CITY+STREET_ADDRESS from ner+regex (S=0.98)
- PIN** (e26): DROP: POSTAL_CODE from regex (S=0.85)
- Shipping address** (e29)
- Vivek Gupta** (e30): DROP: PERSON_NAME from regex (S=0.80)
- you@example.com** (e32): DROP: EMAIL from regex (S=0.95)
- +91 98765 43210** (e34): DROP: PHONE from regex (S=0.90)
- Address** (e36): DROP: STREET_ADDRESS from regex (S=0.95)

A real run on a checkout page. Blue: elements the AI may act on. Red: values removed on the device.

1 Working prototype

Chrome extension, gateway server and three test websites.

2 Runs on an ordinary laptop

Models ship inside the extension. About 1 to 2 s of local work per step.

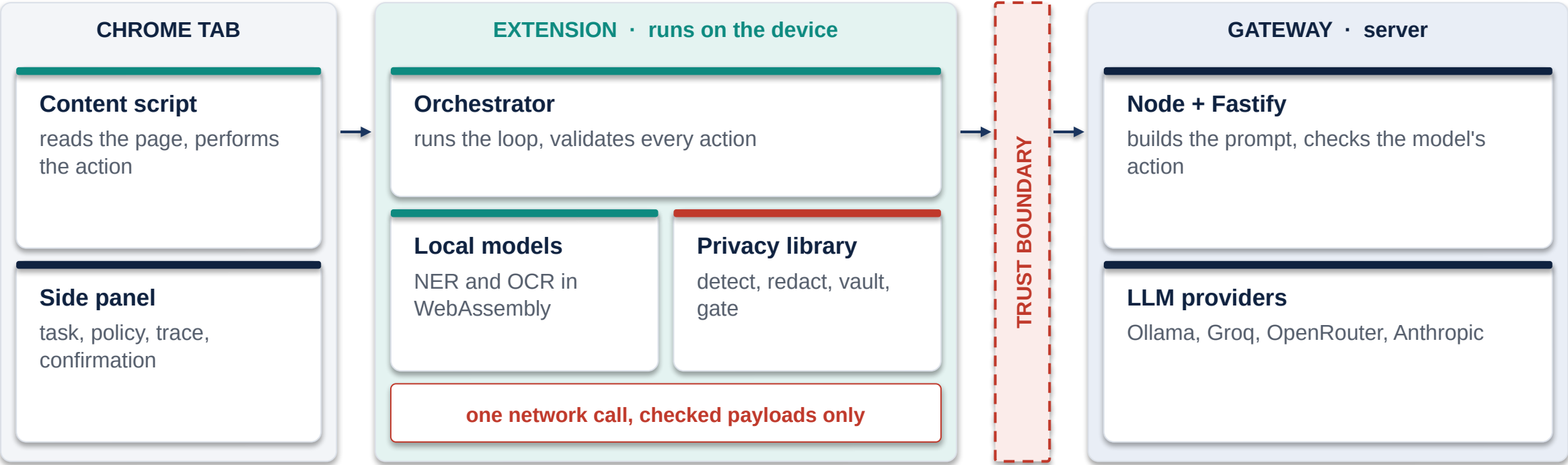
3 Any LLM

Works with a local model (Ollama) or any hosted API. A scripted fallback always answers.

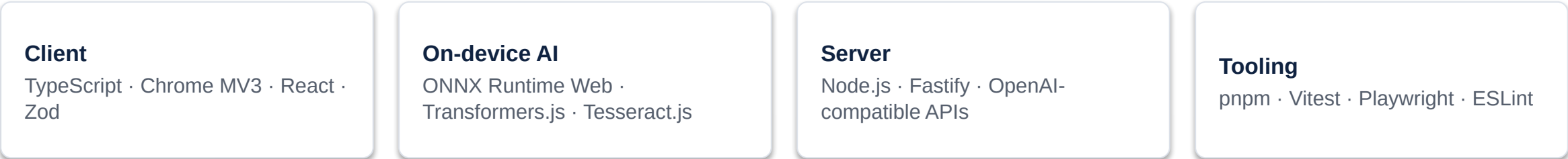
4 Tested end to end

503 unit tests, automated browser runs, and a leak test that can fail.

Architecture and stack



STACK



RESULTS

Measured end to end with the real extension

Page elements found

97.8 %

precision 99.7 %

Personal data caught

100 %

0 leaks

Redaction precision

95.6 %

regions that needed hiding

Local work per step

1.8 s

payload 41 KB

Time per step

2.1 s

p95 2.7 s

Tasks completed

33 / 33

6 tasks · 3 seeds · 2 policies

Automated runs on three test websites, on an ordinary laptop (Apple M1, 8 GB).

WHAT LEAVES THE DEVICE (REAL CAPTURES)

ClinicDesk Seed: 1337

Patient List					
PATIENT ID	NAME	MRN	PHONE	DOB	ACTIONS
P001	Vivek Gupta	CLIN-300178	+917757420286	1998-10-01	View
P002	Vikram Patel	OPD-123733	+919882942278	1956-01-25	View
P003	Ajay Reddy	MRN-243892	+916424795596	1997-07-05	View

Before: the page as rendered.



ClinicDesk

Patient List		
PATIENT ID	NAME	MRN
████	████	████
████	████	████
████	████	████

After: the only image that can leave.

WITH A REAL LLM PLANNER

11 / 11 tasks completed · 0 leaks

3.4 s per step, median

Privacy results did not change; only network and model time was added.

Why it scales, and what comes next

WHY IT SCALES

All perception runs on the user's device, so the server stays small.

Each step sends about 17 to 40 KB, not a full screenshot.

Works with any LLM: local and offline, or hosted.

Same privacy boundary on every website, by design.

NEXT STEPS

A smaller, faster NER model on WebGPU.

A face and document detector for images.

A hosted server with TLS and per-user limits.

Firefox support and a signed audit log.



Thank you

Questions and discussion

GLASSWALL · A privacy-preserving browser agent with on-device visual perception

Source and reports: github.com/rkhoda/GlassWall